Age-based and length-based selectivity

A simulation to inform opal's design, using the IO yellowfin model

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2026-10-14

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1 Why this matters

In the IO yellowfin conversion, opal with SS3′s age-based selectivity held fixed (fit A1) reproduced SS3′s stock size: B0 1.03 × SS3′s and terminal depletion 0.404 against SS3′s 0.399. When opal estimated its own length-based selectivity instead (fits C and C2), B0 fell to 0.88 × SS3′s and terminal depletion to 0.307. The timing simulation showed that catch and biology timing barely move estimated depletion, which leaves selectivity as the main candidate.

SS3′s curves for this model are **age-based**: selectivity is a function of age, and every fish of an age is equally selected whatever its length. opal's are **length-based**: selectivity is a function of length, and reaches age through the distribution of length at age. Two consequences:

1. The same curve family produces different shapes of selectivity-at-age, so one cannot reproduce the other exactly.
2. Length-based selection also changes the predicted length composition *within* an age: larger fish of an age are more likely to be caught. An age-based model predicts the catch's lengths from the population's length distribution at age alone.

This document tests whether these differences bias B0 and depletion, using the model of the timing simulation with opal's timing conventions (start-of-quarter catch and biology) throughout, so that only selectivity differs.

2 Design

2.1 True selectivity

fishery	age truth vs SS3	length approximation of age truth	age approximation of length truth
purse_seine	6e-04	0.0474	0.0139
longline	2e-04	0.0141	0.0090
other	6e-04	0.0712	0.0297

- **Age-based truth:** opal's double-normal curve evaluated on age (the same six-parameter form as SS3's pattern 20), fitted by least squares to SS3's selectivity-at-age for the source fleets of the three pooled fisheries (SS3 fleets 11, 4 and 8). It reproduces SS3's curves almost exactly.
- **Length-based truth:** C2's fitted length-based curves for the same fleets, as in the timing simulation.
- **Cross-approximations:** each truth fitted by least squares in the other basis, on selectivity-at-age. This is the step C2 used to seed opal's curves from SS3's. A length-based curve approximates the age-based truth worse than the reverse: an age-based curve can bend at any age, while a length-based one is smoothed by the spread of length at age.

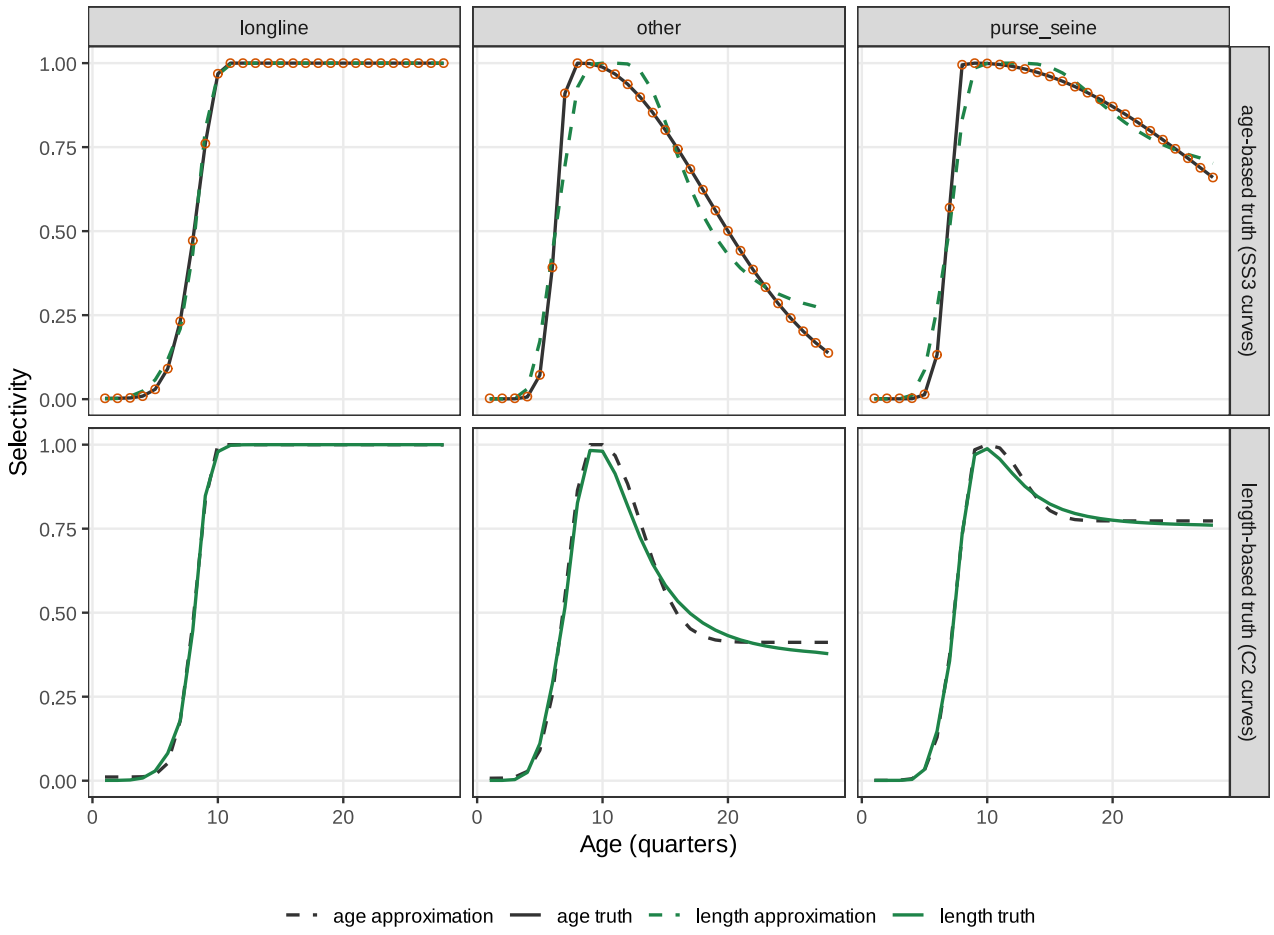


Figure 1: Selectivity-at-age of each truth and its approximation in the other basis, with SS3's curves (orange circles).

2.2 Simulation–estimation

Each of 30 replicates draws recruitment deviations (shared by both operating models) and observation error, with the timing simulation's data: a longline index (CV 0.2, quarters 81–256), multinomial length compositions (ESS 20) for each fishery in every quarter with catch, and catch known exactly. Three estimation models fit each dataset:

Estimation model	Basis	Estimated selectivity columns
`age`	age	1–4 (peak, plateau, ascending and descending width) for all three fisheries, as SS3 estimated
`length (C2 parameters)`	length	C2's choices: 1–4, and 1 and 3 for the longline

Estimation model	Basis	Estimated selectivity columns
`length (all parameters)`	length	1–4 for all fisheries, plus 5–6 (initial and final selectivity) for the two dome-shaped fisheries

All estimate log B0, 256 recruitment deviations (N(0, σ_R) penalty) and log q, and put opal's N(0, 2) prior on the estimated selectivity parameters. Unestimated selectivity columns stay at their starting values: the truth when the basis matches the operating model, otherwise the least-squares approximation, as in C2. Fits start at the true B0, q and recruitment deviations, and are polished with Newton steps.

As in the timing simulation, recruitment-deviation shrinkage biases even the matched models (B0 up, depletion down), so each model is also compared **paired against the matched model on the same dataset**: `age` for the age-based truth, `length (C2 parameters)` for the length-based truth.

3 Part 1: the same parameters, the other basis

Each truth is projected with its approximation in the other basis, with the same B0, recruitment and catch. Nothing is estimated.

Truth	Catch ×	Terminal SSB/B0, truth	with the other basis' approximation	Difference
age-based truth (SS3 curves)	0.25	0.742	0.743	0.2%
age-based truth (SS3 curves)	0.50	0.576	0.579	0.6%
age-based truth (SS3 curves)	0.75	0.414	0.420	1.4%
age-based truth (SS3 curves)	1.00	0.250	0.259	3.7%
age-based truth (SS3 curves)	1.25	0.053	0.072	37.4%
length-based truth (C2 curves)	0.25	0.739	0.740	0.1%
length-based truth (C2 curves)	0.50	0.572	0.573	0.2%
length-based truth (C2 curves)	0.75	0.408	0.410	0.5%
length-based truth (C2 curves)	1.00	0.243	0.246	1.2%
length-based truth (C2 curves)	1.25	0.042	0.048	14.8%

With the same B0, recruitment and catch, swapping a truth for its best approximation in the other basis changes terminal depletion little: at the observed catch, 0.250 for the age-based truth against 0.259 with its length-based approximation (3.7%), and 0.243 against 0.246 for the length-based truth (1.2%). The difference grows with the catch but stays small next to the timing effects. So the shapes alone are not the issue; any larger effect has to come from how the curves are estimated against the length compositions.

4 Part 2: estimating selectivity in each basis

fits	failed	max |gradient| (max)	seconds (median)
180	0	0	17.3295

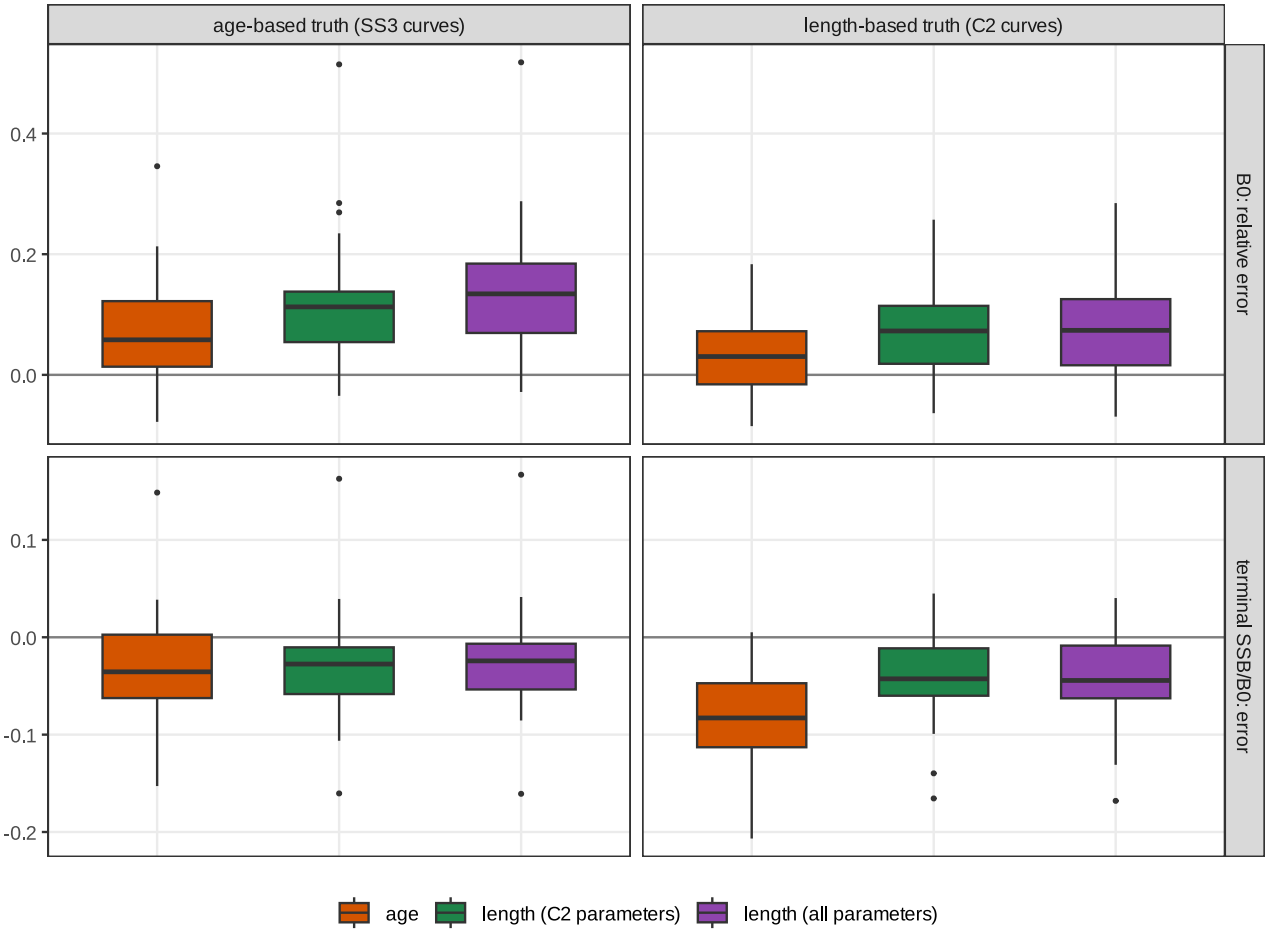


Figure 2: Relative error in B0 (top) and error in terminal depletion (bottom) by estimation model, for each operating model.

Truth	Estimation model	B0 error	B0 vs matched	Term. dep. error	Term. dep. vs matched	IQR vs matched	Mean abs. dep. error
age-based	age *	5.8%	0.0%	−0.035	0.000	0.000 to 0.000	0.066
age-based	length (C2 parameters)	11.3%	4.1%	−0.028	0.001	−0.007 to 0.012	0.065
age-based	length (all parameters)	13.4%	5.7%	−0.024	0.009	−0.003 to 0.019	0.063
length-based	age	3.0%	−4.4%	−0.083	−0.037	−0.046 to −0.029	0.090
length-based	length (C2 parameters) *	7.3%	0.0%	−0.043	0.000	0.000 to 0.000	0.067
length-based	length (all parameters)	7.4%	0.1%	−0.044	0.000	−0.002 to 0.001	0.066

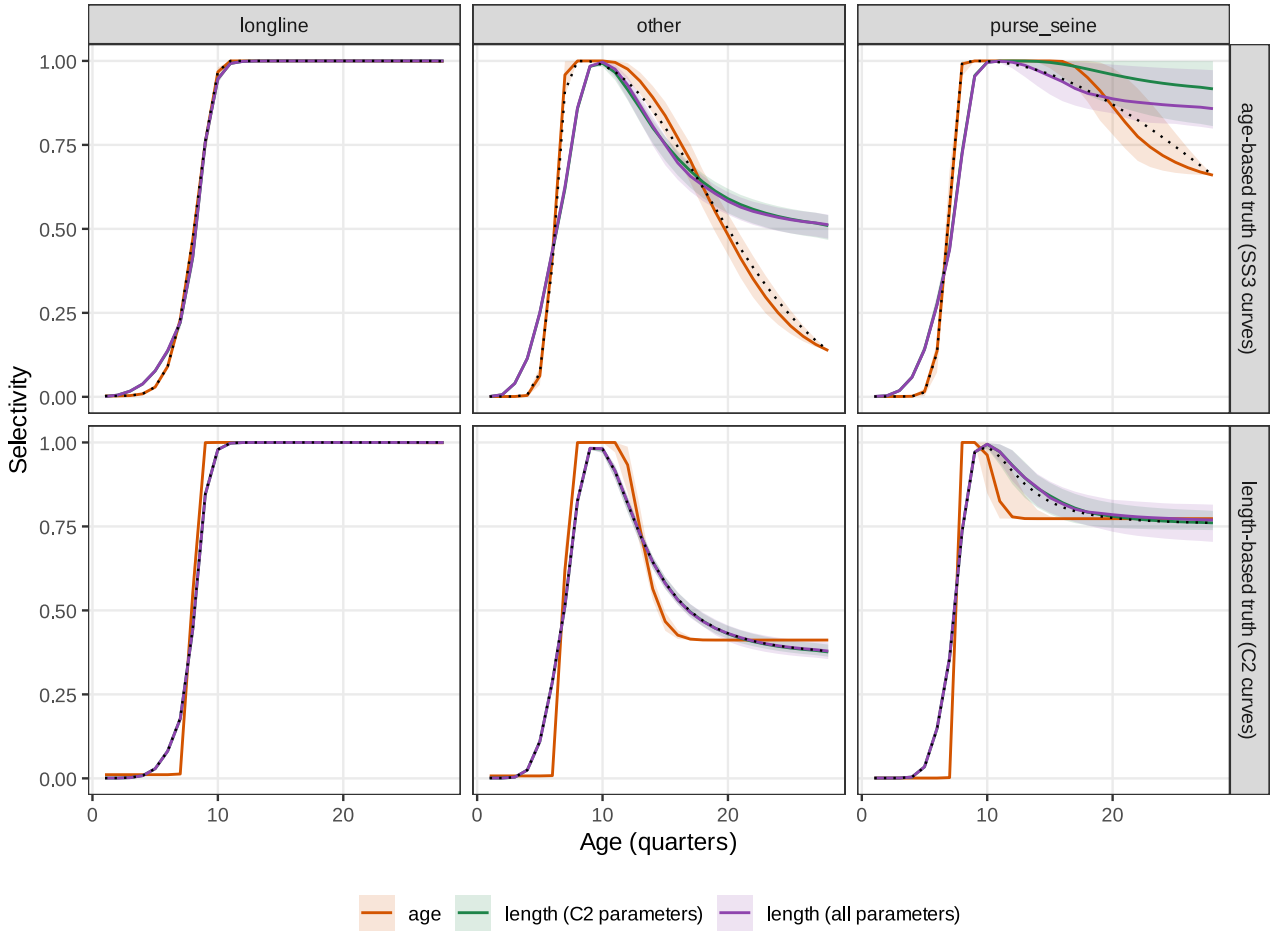


Figure 3: Estimated selectivity-at-age (median and 10–90% range over replicates) against the truth (black).

All 180 fits converged (largest gradient 9.8e-07).

A length-based model fitted to age-based data keeps depletion. With SS3-like age-based selectivity as the truth, the length-based models estimate B0 4.1% (C2 parameters) to 5.7% (all parameters) higher than the matched age-based model, but terminal depletion within 0.001 (10–90%: −0.013 to 0.017). As with timing, the misspecification goes into the scale.

An age-based model fitted to length-based data does not. With opal-like length-based selectivity as the truth, the age-based model estimates terminal depletion 0.037 lower than the matched model (10–90%: −0.058 to −0.024), and B0 4.4% lower: the stock looks more depleted than it is. An age-based model cannot represent the larger fish of each age being caught preferentially, so it misfits the catch's length compositions and absorbs the difference in the stock's trajectory.

Estimating more length-based parameters changes little. Freeing the initial and final selectivity of the dome-shaped fisheries (`length (all parameters)`) moves terminal depletion by 0.000 (length truth) and 0.008 (age truth) relative to C2′s parameter choices.

The direction does not match the IO yellowfin fits. In those fits, opal's length-based selectivity (C2) gave a *lower* B0 (0.88 against 1.03 × SS3′s with SS3′s selectivity) and *lower* depletion (0.307 against 0.404) than SS3′s age-based curves. Here, fitted to the same data, the length-based model's B0 and depletion are at or above the age-based model's in the median: B0 is 4.4% higher and terminal depletion 0.001 (age truth) to 0.037 (length truth) higher. The selectivity basis on its own therefore does not explain C2′s lower stock size.

5 Implications for opal

1. **Length-based selectivity is the safer default.** When the truth is age-based, opal's length-based curves still estimate depletion without bias (B0 is about 4–6% high). When the truth is length-based, an age-based

model biases depletion low. An option for age-based curves would help reproduce SS3 models exactly, but is not needed for unbiased status estimates.

2. **The IO yellowfin gap is not a basis effect.** Both simulations (timing and selectivity) move estimated depletion by at most about 0.04, and in the opposite direction to C2′s lower depletion (0.307 against SS3′s 0.399). The remaining candidates are features these simulations leave out:
   - the full set of 16 fleets, with fleet-specific shapes that a length-based double normal fits poorly (for example fishery 5′s shallow ascending limb and fishery 9′s drop in the plus group);
   - the survey's generalised size-frequency data, which SS3 fitted and the converter omits, leaving the index selectivity weakly informed;
   - opal's N(0, 2) prior on the selectivity parameters, which set nine of C's parameters;
   - data from a spatial operating model, where pooled area fleets need not follow either basis. A next step would be to rerun this design with all 16 fleets, or to refit C2 with the survey's size-frequency data, to see which of these moves depletion by 0.1.
3. **Freeing more selectivity parameters is not the fix.** Estimating the initial and final selectivity of the domes made almost no difference.

6 Reproducing

Step	Script	Does	Time
1	`01-om-inputs.r`	True and approximating selectivity curves; reuses `../timing/data/om-inputs.rds`	~10 s
2	`02-deterministic.r`	Part 1 projections	~10 s
3	`03-sim-est.r [n_rep] [n_cores]`	Part 2, 30 replicates × 6 fits	16 min on 4 cores
4	`selectivity-sim.qmd`	This document; reads only `data/`	seconds

The model is the timing simulation's (`../timing/R/timing-model.R`), which gained a `sel_basis` argument for this study. Results were generated 2026-10-14 08:33:13 UTC.